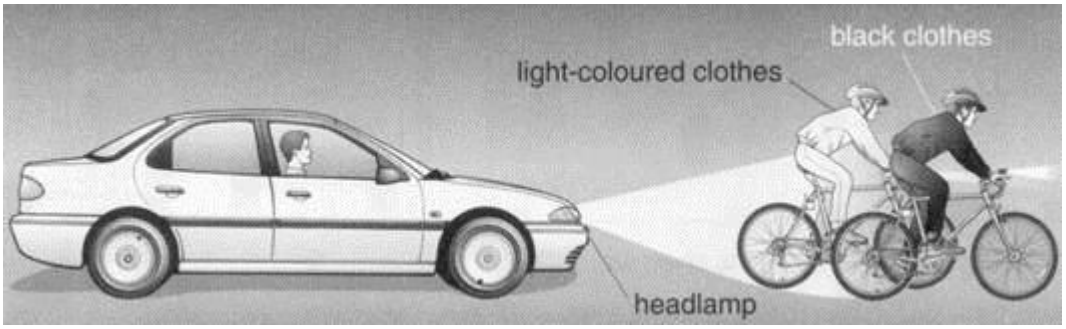
Q1.

Two cyclists are riding along a dark road at night. One is wearing black clothes and the other is wearing light-coloured clothes.



A car is driving behind the two cyclists. Light from the car headlamp shines on the cyclists.

(a) What happens to the light when it reaches the light-coloured clothes?

…….

1 mark

(b) **On the drawing above**, draw a ray of light to show how light from the headlamp reaches the driver so that he can see the cyclist in the light-coloured clothes. Draw arrows to show the direction of the light.

3 marks

(c) What happens to the light when it reaches the black clothes?

…….

1 mark

Maximum 5 marks

Q1.

(a) it is reflected

accept 'it is scattered'

*accept 'it reflects **or** bounces off'*

1 (L5)

(b) one mark is for a ray from the headlamp to the light-coloured clothes and from them to the driver's eye

***both** parts of the ray are required*

accept small discontinuities in the ray

accept rays which are almost straight but which have not been drawn with a ruler

*do **not** accept dotted lines*

1 (L5)

- one mark is for accurately drawing the correct ray

*this is a dependent mark do **not** award this mark unless the first mark was also awarded*
the ray must touch the headlamp, touch the cyclist's clothes, and touch the driver's eye the ray must be continuous
both parts of the ray must be drawn with a ruler

1 (L5)

- an arrow showing that the light enters the eye **or** leaves the headlamp

accept a correct arrow on an incomplete line

1 (L5)

(c) it is absorbed

accept 'it absorbs'

accept 'most is absorbed'

accept 'it is not reflected'

accept 'only some is reflected'

1 (L5)

[5]